Microsoft

Microsoft Education AI Toolkit

A navigator for education institutions to plan their AI journey

The power of the possible: The promise of generative AI in education

In today's educational landscape from primary, secondary and higher education institutions, generative AI is emerging as a powerful tool that can change the way we teach and learn. For educational leaders, generative AI's potential is both vast and inspiring, offering significant opportunities to enhance educational outcomes, foster creativity and, most importantly, prepare students for a future where AI is an integral part of every work and life.

Generative AI tools are versatile and capable of supporting a wide range of educational modalities. From personalised learning experiences to collaborative projects, generative AI will be able to adapt to each student's unique needs, offering tailored support to help maximise individual potential. Imagine classrooms where an educator's lesson plans are dynamically customised to suit the pace and style of each learner, or higher education institutions where virtual tutors provide instant feedback and support advanced research endeavours. These AI-driven experiences are starting to become a reality in some educational environments.

The impact of generative AI is already being felt. In primary and secondary education, educators use AI tools to create tailored simulations, develop engaging educational activities and produce interactive storytelling experiences that captivate students. In higher education, generative AI is enhancing research capabilities, enabling students and faculty to analyse vast datasets and generate new insights with unprecedented speed and accuracy. Early users show improved student engagement and achievement, as AI streamlines administrative tasks and lesson planning, helping to enable educators to focus on individual student needs. IT departments benefit from AI's speed in protecting data, and students gain valuable skills for an AI-driven future. These advancements are making education more efficient, personalised and aligned with future job market demands.

While the potential benefits of generative AI are significant across all levels of education, addressing concerns such as data privacy, algorithmic bias and the need for human interaction in learning is also essential. Ensuring that AI systems are used responsibly is imperative. By integrating AI in ways that complement rather than replace human educators and focusing on data security and fairness, we can build trust and create a more effective and inclusive educational environment.

The introduction of generative AI into education is more than a technological upgrade; it is a call to inspire the next generation of innovators. As educational leaders whether guiding young minds or shaping future professionals, you have the opportunity to harness generative AI to ignite curiosity, foster creativity and cultivate a lifelong love of learning in your students. By integrating these tools, you're not only enhancing the educational experience, but also equipping students with the skills and mindset they need to thrive in an ever-changing world.

We invite you to explore our comprehensive toolkit, designed to provide you with the knowledge and resources needed to implement generative AI effectively in your educational setting.

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A brief overview of AI

Artificial intelligence emerged in the 1950s when early pioneers like Alan Turing laid the groundwork for machine intelligence, posing the question of whether machines could exhibit human-like thinking. In 1956, researchers met at the Dartmouth Conference to explore the idea of creating machines that could think like humans – this is considered the beginning of AI.

Throughout the following decades, AI garnered periods of enthusiasm and scepticism. However, as computers have become more powerful and we have more data and better algorithms, AI has advanced, especially the subset of AI called machine learning (ML). Neural networks and deep learning techniques have made ML more powerful and useful. Advancements in natural language processing (NLP) have enabled machines to understand, interpret and generate human language making AI even more accessible to a diverse group of users.

Today, AI is used in many parts of our lives, from virtual assistants and recommendation systems to autonomous vehicles and diagnostic tools. Smart home devices, email filtering and language translation apps also use AI. An area of significant advancement in AI technology is generative AI. Generative AI allows any user to prompt the tool to create text, images, code snippets and more. What sets generative AI apart is that it makes creating easy for nearly everyone, even if they don't have special technical skills. It offers a simple way for all users to make their own content.

Small language models (SLMs), a recent innovation within AI, offer a compact alternative to large language models (LLMs) like ChatGPT. SLMs are designed to understand, generate and interpret human language using fewer resources, which makes them particularly beneficial in K-12 and higher education settings. Despite their smaller size and training on less data compared to LLMs, SLMs can perform many of the same tasks, such as language processing, coding and basic maths. These models require less computational power and memory, making them ideal for deployment on mobile devices and in resource-constrained environments often found in educational institutions.

SLMs are well-suited for applications that need to run locally on a device, offering benefits like reduced latency and increased privacy – important considerations in educational environments. Additionally, they are ideal for tasks that don't require extensive reasoning or complex data analysis, making them especially useful in regulated industries and areas with limited network access. While SLMs may have a more limited knowledge base and a narrower understanding of language and context, their efficiency enables schools to integrate AI into a broader range of applications, providing scalable solutions that support personalised learning experiences and administrative efficiency without the need for extensive infrastructure.

It's all about the data

Data is, in many ways, at the centre of educational institutions, playing a pivotal role in shaping strategies, enhancing teaching methodologies and fostering continuous improvement. Starting the AI journey in education requires a fundamental shift in perspective: recognising that thoughtful consideration of data management is central to the strategic planning needed to leverage the full potential of AI technologies. This isn't just about technology; it's about shaping a future where strategic decisions are informed and enhanced by AI.

One of the more challenging barriers to implementing AI solutions in education is the existence of data silos, separate information repositories confined within various proprietary software systems. These silos not only hinder access to data but also impede the holistic analysis necessary for data-driven decision-making.

Addressing this challenge requires a concerted effort to dismantle data silos and embrace a more inclusive approach to data aggregation. By breaking down these barriers and adopting a unified data approach, schools can achieve deeper insights and offer personalised learning experiences, enhancing AI's impact. This not only enhances the capability of AI systems to generate meaningful outcomes but also significantly advances the institution's ability to meet the evolving needs of its students and educators.

Initially, the focus should be on establishing a foundational data management strategy that begins to break down silos and integrate diverse data types. Simultaneously, implementing basic security measures to protect this data is essential. As institutions evolve, so too can their data management and security practices, evolving from simple, initial set-ups to more sophisticated systems like data lakes and advanced encryption methods.

This approach emphasises the importance of starting where you are, with what you have and understanding that perfection is not a prerequisite for progress. Incremental improvements, fuelled by ongoing learning and adaptation, are key to building a robust framework capable of supporting the sophisticated needs of AI technologies. Encouraging a culture of continuous improvement and shared responsibility among all stakeholders – administrators, IT staff, faculty and students – can significantly enhance both data management practices and security postures.

Microsoft's integrated cloud platform, Microsoft Azure, offers a comprehensive solution. Azure simplifies the development of advanced solutions and the use of analytics. Educational institutions using Microsoft cloud solutions can centralise their data management, increase operational flexibility and provide personalised experiences for students. This solution isn't just about technology; it's about shaping a future where strategic decisions are informed and enhanced by AI. As a result, data-driven decisions informed by AI constitute a fundamental shift from storing, accessing and analysing data to dialoguing and conversing with your data.

Quality and diversity of data over volume

The true power of AI is unlocked not by the sheer volume of data, but by its diversity. Educational institutions have access to a wide range of data types, including academic records, multimedia and behavioural metrics. However, the integration of these varied data sources is crucial to harness their full potential. For AI systems to function well, they require a broad spectrum of data types. The key to unlocking valuable insights from AI lies in the variety and volume of data. This diversity enables AI to generate more comprehensive insights. The concept of big data is key here – it's not just about large amounts of data, but about varied and comprehensive datasets that feed into large language models, significantly enhancing their performance.

Small language models (SLMs) further complement this by being adaptable to institutions that may not have extensive datasets or the infrastructure to support large-scale AI. Their efficiency and lower resource demands make SLMs an accessible entry point for schools and universities looking to integrate AI-driven solutions without overhauling existing systems.

The journey toward effective AI integration in education is marked by learning from challenges, celebrating advancements and remaining adaptable to the ever-changing digital landscape. Each step forward, no matter how small, contributes to the overarching goal of transforming educational environments through the power of AI, underpinned by solid data management and security foundations.

Information lifecycle and governance in the age of AI and storage limits

Weak information governance exposes organisations to risk and undermines generative AI adoption. In this recorded webinar, hear from Gartner analyst, Max Goss and Microsoft on how this impacts education institutions. This discussion provides practical guidance on how to more effectively manage the information lifecycle to both meet new storage parameters and prepare for the future of AI.

AI and the future of work

Artificial intelligence, including generative AI, is significantly transforming the landscape of work, necessitating a re-evaluation of the skills required in the workforce. As AI integrates into various industries, it creates new opportunities that require a blend of technical proficiency and durable skills like critical thinking and emotional intelligence. Recognising these shifts, reports from professional organisations underscore the urgency for an updated skills framework, highlighting a growing demand for a workforce adept at navigating a technology-driven environment.

> "If our era is the next Industrial Revolution, as many claim, AI is surely one of its driving forces."
>
> Dr. Fei-Fei Li
> Stanford University

This evolution in the job market calls for a corresponding shift in educational models. Traditional curricula, often criticised for their slow adaptation to technological advancements, must transition towards more dynamic, personalised learning experiences that foster technical literacy and cultivate the skills essential for the AI era such as curiosity and metacognition. These skills are essential for understanding AI and harnessing its potential. Proficiency in prompt design is also crucial for effective information retrieval and content creation.

The path forward involves strategic planning that integrates AI and future skills into curricula, fostering partnerships with technology companies and creating a culture that values innovation and adaptability.

Our evolving educational landscape emphasises the shift from creating content to analysing and integrating skills, driven by AI's capacity to rapidly locate and generate content. However, it's important to recognise that generative AI isn't perfect and can sometimes produce inaccurate information or fabrications. Students must be adept at identifying errors and inaccuracies.

To succeed in an AI collaborative world, students, educators and perhaps every member of an educational community must adapt or learn new skills like how to prioritise, delegate, proofread, review and master efficiency. Equipping students and educators with these skills is essential for success in an AI collaborative world. Discover the future of work with WorkLab, a Microsoft platform offering science-based insights, expert advice and compelling stories about AI. Stay ahead with articles, reports and podcasts that reveal how technology is transforming the workplace.

Copilot for education leaders

The role of an education leader is more than just managing daily operations. Leaders shape and enact policies, make data-based decisions, monitor achievement, implement curricula and oversee faculty development. Microsoft Copilot helps education leaders accomplish many of these time-consuming tasks.

Microsoft Copilot

Increase productivity when completing administrative duties to:

- Research and compare curricula
- Outline an agenda for professional learning
- Summarise online articles or PDFs

Microsoft 365 Copilot

Use Microsoft 365 apps and files to complete specialised tasks to:

- Summarise internal state reports
- Auto-draft messages to faculty
- Create visualisations from spreadsheets

Copilot prompt

Summarise the *2024 National Educational Technology Plan* and Include sections on the digital use divide, digital design divide and the digital access divide. Provide a one or two sentence definition of each digital divide and list five steps to take to address the divide in each section. The summary should be written in plain language that's understandable by educators. Cite any source material.

Copilot for educators

As the people most directly responsible for student learning, educators spend the bulk of their working hours writing lesson plans, assessing understanding, facilitating classroom activities and completing administrative duties. Microsoft Copilot makes common educator tasks more manageable and efficient.

Microsoft Copilot

Increase productivity and save time completing duties to:

- Create a course syllabus
- Write a lesson plan that differentiates instruction
- Level text for emergent readers

Microsoft 365 Copilot

Use Microsoft 365 apps and files to accomplish specialised tasks to:

- Recap Teams meetings for absent students
- Auto-draft emails for families
- Create a rubric from a lesson document

GitHub Copilot

Deploy an AI-powered coding assistant that supports computer science instruction to:

- Provide students with just-in-time coding support
- Debug complicated programs and refactor code
- Help students document change logs

Copilot prompt

You are an AI with expertise in physics. Your task is to provide five diverse analogies that can help explain Bernoulli's Principle to high school students preparing for their state exams. The analogies should be simple, concise and cater to a range of student interests and experiences. Remember, your goal is to aid their understanding of the principle, not to introduce more complexity.

AI for students

By equipping students with the knowledge and tools needed to safely interact with AI products in the classroom, we can better prepare them for the real-world challenges and future workplaces that they will encounter. Recent research, sponsored by Microsoft, reveals significant insights into the widespread adoption of AI in schools. Explore the key findings.

- 35% of students use AI to summarise information, the highest usage for students.
- Microsoft Research and Harsh Kumar of the University of Toronto discovered that AI-generated explanations enhanced learning compared to solely viewing correct answers.
- Harvard University and Yale University professors found that AI chatbots can give students in large classes an experience that approximates an ideal one-to-one relationship between educator and student.

Minecraft Education

Minecraft Education offers a set of accessible, engaging materials for building AI literacy with Minecraft. For instance, the AI Foundations program is designed to empower students, educators and families with a fundamental understanding of how AI works and how to use AI tools responsibly. Students can learn the basics of AI, how AI helps solve problems and ways to be responsible when using AI tools with the AI Adventurers video series.

Explore these Minecraft Education worlds to get started

Experience	Description	Age
Fantastic Fairgrounds	Discover how to unlock the power of AI through a wondrous world! Practice the skills to understand, evaluate and utilise this exciting technology.	**Ages 8-18**
Hour of Code: Generation AI	Develop problem solving, creativity and computational thinking skills along AI foundations. Learn the basics of coding in MakeCode Blocks or Python and help build better AI for all.	**All ages**
AI for Earth	Use the power of AI in a range of exciting real-world scenarios, including preservation of wildlife and ecosystems, helping people in remote areas and research on climate change.	**Ages 8-18**

Classroom AI Toolkit

The Classroom toolkit: Unlocking generative AI safely and responsibly is a creative resource that blends engaging narrative stories with instructional information to create an immersive and effective learning experience for educators and students aged 13 to 15 years.

With the toolkit, educators can initiate important conversations about responsible AI practices in the classroom with their students. By completing the lessons, students gain valuable insights and develop practical skills to enhance their digital safety.

These simple tips can help your students successfully use Copilot and other generative AI tools.

- **AI as a Copilot:** Think of generative AI tools as your helpful assistants. They follow your commands and perform tasks well, but it's up to you to use them wisely and responsibly.
- **AI is not perfect:** While AI tools can do a lot of things well, these tools can make mistakes because they are trained to always provide an answer. This makes it important to stay alert.
- **Always fact-check:** Make fact-checking a habit. Do not blindly trust AI-generated information – always verify it with trusted sources to be sure.
- **Beware of bias:** Generative AI models can sometimes show bias in their responses. Always ensure you review the outputs with a critical eye and be proactive by adjusting the prompts as necessary.
- **Always cite your sources:** Ensure that you give credit where it is due by always citing work that has been completed with the support of generative AI.
- **Protect your information:** Don't share private information with untrusted websites or apps and read privacy policies to understand how your data is used. Don't forget you can use AI tools to summarise complex documents, but always remember to fact-check and verify!
- **Mind your well-being:** Communicating with an AI tool that can appear to converse naturally with you can be very tricky. Establish healthy boundaries with technology by limiting screen time and spending time with the important people in your life.

Use AI-powered tools and prepare for AI

Using their school-issued Microsoft accounts, students have access to AI-powered Learning Accelerators at no cost. This commitment to accessibility and equity ensures that all students, regardless of background or financial means, can leverage cutting-edge technology to enhance their educational journey.

AI tools

		Students 18+	Students 13+	All Students
	Microsoft Copilot	●	*	
	Microsoft 365 Copilot	●		
	GitHub Copilot	●	*	
Learning Accelerators	Reading Coach	●	●	●
Learning Accelerators	Search Coach	●	●	●

* Currently in private preview.

Microsoft Copilot

When students use Copilot, they immediately gain access to an on-demand AI assistant that can help provide contextualised explanations of challenging concepts, brainstorm creative project ideas and offer instant feedback on assignments.

Learning Accelerators

Microsoft's Learning Accelerators offer AI-powered support to help students enhance their literacy, maths, social-emotional, speaking and information literacy skills. These tools provide personalised coaching, immediate feedback and practical exercises. Notable examples include Reading Coach and Search Coach. When used alongside direct instruction and guidance from educators, these tools help primary and secondary students develop essential skills in reading, writing, listening and speaking.

Reading Coach provides personalised, engaging AI-powered reading fluency practice for all learners. This powerful tool helps students develop literacy skills by earning achievements and unlocking new story elements as they read, driving continuous engagement. The AI features in Reading Coach include AI-generated stories and personalised literacy practice.

Search Coach was designed with the help of leading information literacy experts to provide tips and strategies for students as they search for and evaluate sources of information.

Search Coach helps students prepare for AI and enhance their information literacy skills by customising filters to refine their search queries to find the most credible and relevant information.

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Learning Accelerators

Get to know Microsoft Learning Accelerators

Reading Progress
Tracks student reading skills and provides educators with actionable insights for targeted improvement areas.

Reading Coach
Offers AI-powered, personalised reading fluency practice, enabling learners to co-create stories and practice challenging words.

Search Progress
Enables educators to guide students' information literacy skills by monitoring their search activity and query quality.

Search Coach
Fosters information literacy by coaching students to develop effective search queries and identify reliable resources.

Speaker Progress
Provides data-driven insights on students' speaking skills.

Speaker Coach
Offers real-time feedback on public speaking elements within PowerPoint and Teams.

Reflect
Encourages students to identify and express emotions and provides educators with insights to offer support.

Maths Progress*
Aids educators in creating practice questions and analysing students' challenges, facilitating personalised feedback and support.

Maths Coach**
Enhances maths learning with real-time feedback and personalised practice for students.

Education Insights
Integrates data across Learning Accelerators to equip educators with a comprehensive view of each student's academic journey.

* Entering private preview

** Coming soon

AI Snapshot

Improving efficiency in K-12 grant writing

How Microsoft Copilot and Microsoft 365 Copilot can support grant identification and streamline the application process

K-12 Grant Coordinator

Use generative AI to find and apply to more grants and improve the efficiency of the process.

Goal

Improve efficiency and productivity in identifying and preparing grant applications.

Technology

- Copilot
- Microsoft 365 Copilot

1. Visit Microsoft Copilot.
   Note: Be sure you're signed in using your school account to ensure enterprise data protection is enabled. Additionally, ensure that 'Web' is selected for the following prompts. That setting will ensure only publicly available information is accessed and not private data on your PC.
2. Copy-paste the following prompt into Copilot and update the highlighted text to reflect the name of your school district or organisation:
   Analyse available information about my school district, ***[School or Organisation]*** *and identify five key needs that could be addressed by a publicly available grant.*
3. Continue the conversation with Copilot by asking it:
   Are there specific grant programmes available for the first item on the list for my school district?
4. Next, ask Copilot:
   Craft a clever title for the grant application and then draft an outline to apply for the first grant programme you've identified. For each point in the outline, include a 300-word response that addresses the grant application requirements. Be sure to include citations for each section.
5. Use the copy icon in Copilot (or the Export to Word option) after the response in step 4 to copy Copilot's response and paste it into a new Word document.
6. Open the Copilot sidebar in the Word document, select one of the sections of the outline that needs additional information and enter the following prompt:
   What indices should I include in this section?
   Note: You may have to copy-paste the excerpt into Copilot.
7. Use this outline and summary as the starting point of a grant application. You may also want to use Microsoft 365 Copilot in Word to prompt for additional data or justification for the potential grant application.

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AI Snapshot

Transforming student engagement with relatable and relevant content

How Microsoft Copilot can help educators increase accessibility by making student learning more engaging, and relevant, ultimately boosting outcomes for all students

K-12 Educator

Use Copilot to generate relevant examples when explaining new concepts, making the content more relatable and easier to understand for your students.

Goal

Help ensure directions and explanations are accessible for all students.

Technology

Microsoft Copilot

1. Visit Microsoft Copilot.
2. **Note:** Be sure you're signed in using your school account to ensure enterprise data protection is enabled. Additionally, ensure that 'Web' is selected for the following query.
3. Copy-paste one of these prompt ideas into Copilot. Tailor any relevant information to your needs.
   - **a.** I'm teaching a lesson on ecosystems to English Language Learners (ELL) students from Mexican, Vietnamese and Somali backgrounds. Can you provide an example of a food chain that includes animals relevant to these cultures?
   - **b.** I'm teaching the Pythagorean theorem to 9th-grade students with interests in basketball, guitar playing and video game design. Can you provide an explanation of the theorem tailored to each of these interests?
   - **c.** I'm teaching [concept] to [audience] students with [backgrounds/interests]. Can you provide an explanation of [concept] tailored to each of these interests?
4. Copilot will generate the examples, but don't leave Copilot yet. Copilot can continue the conversation and go deeper. Try asking Copilot:
   - **a.** Can you create a quiz question using each of these examples to check for understanding?
   - **b.** What are some common misunderstandings students have about this concept?
   - **c.** What is a hands-on activity we could do to help solidify the learning?
   - **d.** What other real-world contexts could we explore where [concept] is used?
5. When you're finished, export your responses to a Word document, PDF or text file to share with your students, or copy and paste them to another location for easy access.

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Improving cybersecurity with custom cybersecurity promptbooks

How Copilot for Security can improve cybersecurity for technology through collaborative and custom prompts

IT Cybersecurity Specialist

Achieve consistent expert-level analysis and comprehensive reports across the IT team by creating and sharing custom Copilot for Security promptbooks.

Goal

Improve cybersecurity against evolving threats and vulnerabilities using custom AI prompts tools that save IT admin time and improve protection.

Technology

Copilot for Security

1. Access Copilot for Security.
   a. Access your Azure portal.
   b. Search for and select Copilot for Security.
   Note: Microsoft Copilot for Security is offered on a consumption-based model on the number of Security Compute Units (SCU) used.

2. Create a promptbook.
   a. Type a question for Copilot for Security and select ‘Send’ or ‘Enter’. Use this sample prompt to get started:
   If a student is listed in the incident details, show which devices they recently used and indicate if they are compliant with policies.
   b. Select the checkboxes beside the prompts to include them or select the top box to include all prompts in the session.
   c. Select Create to create your new promptbook.
   d. Test your promptbook by selecting the View icon.

3. Share a promptbook.
   a. Go to the Promptbook library in the main menu and look for your promptbook.
   b. Select . . . , then select Details from the options.
   c. Review the pre-built Promptbook Library.
   d. Select Share to get a link to the promptbook that you can share with other users in your organisation.
   e. Learn more about effective prompting.

AI Snapshot

Supporting students with AI-driven insights

How Microsoft Fabric empowers academic leaders to support students through AI-driven data insights

Dean of Students

Analyse student engagement to pinpoint challenges and employ AI-driven recommendations for targeted interventions aimed at supporting student persistence.

Goal

Improve identification and support for struggling students through enhanced data analysis.

Technology

- Fabric
- Copilot for Fabric

1. Access Fabric.
   a. Open the Fabric homepage and select the **Account manager**.
   b. In the Account manager, select **Start trial**. If you don't see the Start trial button, trials might be disabled for your tenant.
   c. Use the Admin centre **Capacity settings**. All users with access to those workspaces are now able to use that trial capacity. The Fabric administrator can edit **Capacity settings** as well.

2. Set up a Task Flow.
   a. Navigate to the workspace where you want to create your task flow and open **List view**.
   b. Select a pre-designed task flow on the empty default task flow, by choosing **Select a task flow**.
   c. Add a new task to the task flow canvas, open the **Add dropdown** menu and select the desired task type.
   d. Edit the task name and description.
   e. Change the task by opening the task details pane and then selecting from the **Task type** dropdown menu.
   f. Arrange the tasks by selecting and dragging each task to the desired position in the task flow.
   g. Add connections by selecting the edge of the starting task and drag to an edge of the next task.

3. Assign items to a new task.
   a. Once a task has been placed on the canvas, assign items to it to help structure and organise the work. Create new items to be assign to the task, or assign items that already exist in the workspace.

4. Enable Copilot for Fabric.
   a. Copilot and other generative AI features in preview bring new ways to transform and analyse data, generate insights and create visualisations and reports in Microsoft Fabric and Power BI.

AI Snapshot

Sparking your students' curiosity with AI-powered teaching assistants

How Khanmigo for Teachers helps educators make relevant instructional content that connects to students' interests

Elementary teacher

Connect lesson topics with real-world context and students' lives to boost engagement and relevance.

Goal

Make learning materials more meaningful and accessible through relevant connections for students.

Technology

Khanmigo for Teachers

1. Access Khanmigo for Teachers.
   Note: Khanmigo for Teachers is available for free in 40+ countries in partnership with Microsoft.
   a. Go to khanmigo.ai/teachers.
   b. Select **Teacher**.
   c. Choose an option for creating an account.
   d. Fill in the required information on the form.
   e. Select **Sign up**.

2. Generate content with real-world context.
   a. From the Khanmigo for Teachers homepage, select **Real World Context Generator**.
   b. Set the grade level.
   c. Add the instructional topic and then select **Write some ideas**.
   d. Review and customise the generated content.

3. Connect content to students' passions.
   a. From the Khanmigo for Teachers homepage, select **Make it Relevant**.
   b. Add the learning objectives.
   c. Add students' interests and then select **Make it relevant**.
   d. Review and customise the generated content.

4. Customise content.
   a. Highlight a word or passage from the generated text.
   b. Select from the following options in the pop-up menu:
      i. Make changes to this: Offer Khanmigo direction such as, "Turn this into a five-minute station activity."
      ii. Try something new: Request an entirely new option without needing to add any directions.
      iii. Discuss this: Open a side-bar discussion with Khanmigo.

AI Snapshot

Automating transcripts and redactions

How Microsoft 365 Copilot in Teams increases efficiency and protects sensitive information

Administrative assistant

Summarise key discussion points – including who said what and where people are aligned or disagreed – and suggest action items, all in real time during a meeting.

Goal

Keep teams connected and productive with real time meeting summaries that automatically assign action items and protect sensitive information.

Technology

Microsoft 365 Copilot in Teams

1. Access Microsoft 365 Copilot in Teams.
   a. Assign Microsoft 365 Copilot add-on licences for intended users.
   b. Open the **Teams** admin centre.
   c. Expand **Meetings** from the navigation pane.
   d. Under **Meetings**, select **Meeting Policies**.
   e. Either select an existing policy or create a new one.
   f. Select **On** or **On only with retained transcript** from the dropdown for the Copilot setting.
   g. Select **Save**.

2. Improve efficiency during Teams meetings.
   a. Select the Copilot icon in from the toolbar.
   b. Chat with the Copilot assistant using these suggested prompts:
      i. What are some follow-up questions that I can ask in an email?
      ii. Create a table with the ideas discussed and their pros and cons.
   c. Select **More prompts** and choose from the following:
      i. Recap the meeting so far.
      ii. List action items for each person.
      iii. Generate meeting notes.
   d. Close a meeting.
      i. Copilot will send a prompt a few minutes before a meeting's scheduled end to help participants wrap up.
      ii. Select **Open Copilot** to see a summary of key points of discussion and identify agreed-upon next steps, including tasks assigned to specific people.

3. Follow-up after a Teams meeting.
   a. From the meeting chat, go to the Recap tab and open Copilot. From here, Copilot bases responses on the meeting transcript.
   b. Try these prompts. Copy them or modify them to suit your needs.
      i. Draft an email to the meeting participants that summarised the meeting and includes the action items. Redact any sensitive information.
      ii. What questions were asked, answered and unresolved?
      iii. Summarise what people said, in a less technical way.

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Section 2

AI Navigators

A global collection of best practices

Education AI Navigators

Microsoft is excited to share the stories of institutions leading the way with research, experimentation, testing and deployment of generative AI solutions in education – the AI Navigators. These trailblazers span a range of countries and educational organisations – from Ministries and State Departments of Education to institutions of higher education as well as primary and secondary schools. Each story highlights a common theme including:

- **Improved cybersecurity**
- **Greater efficiency**
- **Insights from data**
- **Increased accessibility**

Microsoft Education AI Navigators are leading the way in establishing best practices that unlock the full potential of generative AI. Use these materials to follow in their footsteps to assess your organisation’s AI readiness, acquire the necessary technology and take the first steps toward building your own AI capability using their implementations as your guide.

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Insights from data	[California State University, San Marcos](#) Higher education	Dynamics 365
	[New York City Public Schools](#) K-12	Azure OpenAI Service
Increased accessibility	Auburn University Higher education	Microsoft 365 Copilot
	The Education University of Hong Kong Jockey Club Primary school	Azure OpenAI Service
	University of South Florida Higher education	Microsoft Copilot
	Washington State Office of the Superintendent of Public Instruction State Department of Education	Microsoft Copilot
Improved cybersecurity	Department for Education, South Australia Ministry of Education	Azure AI Studio
	Oregon State University Higher education	Copilot for Security
Greater efficiency	Indonesia Ministry of Education and Culture Ministry of Education	GitHub Copilot
	Sikshana Foundation Education foundation	Azure OpenAI Service
	University of Sydney Higher education	Azure OpenAI Service
	Wichita Public Schools K-12	Microsoft Copilot

Auburn University

A higher education institution built a culture of innovation through the responsible use of AI.

Visit website

Increased accessibility

Greater efficiency

To enhance research and learning outcomes, Auburn University in the U.S. state of Alabama integrated Microsoft Copilot, Microsoft 365 Copilot and Azure OpenAI Service into its academic framework. Auburn fosters a culture of innovation by empowering students and faculty to explore creative, AI-driven solutions across disciplines, while promoting AI literacy, secure and responsible usage and collaboration to prepare their community for future advancements.

After extensive stakeholder engagement, Auburn developed a course to boost AI literacy and support learning. They offer classes and workshops on building chatbots, applying AI in business and more. Auburn is also testing Microsoft 365 Copilot with 100 faculty members to improve efficiency and hosted an 'AI Day' with over 400 attendees, featuring discussions on AI integration, safeguards and future possibilities.

- How do your current needs align to the driving forces behind Auburn's AI story? What does responsible use of AI mean to you for staff and students?
- How might using Microsoft Copilot empower your faculty and students to explore AI-driven innovation?
- How might you develop a common understanding of AI literacy across your institution? What training and support might you need to put in place to support AI literacy?

"Our goal is to democratise the value of AI. The focus extends beyond the efficiencies of AI authoring. It's about equipping our Auburn community with the ability to apply AI in creative and ethical ways, integrating it into our daily fabric as seamlessly as mobile phones have over the past decade."

– John Davidson,
Assistant Vice President and Chief Technology Officer, Auburn

AI Tool: Microsoft Copilot

The Education University of Hong Kong Jockey Club Primary School

Visit website

A primary school reimagined teaching and learning with GenAI chatbots using Microsoft Azure OpenAI.

Increased accessibility

The Education University of Hong Kong Jockey Club Primary School (EdUJCPS) created chatbots using Microsoft Azure OpenAI Service to create a more engaging, personalised and secure learning environment so educators can focus on instructional strategy, using AI to provide real-time feedback and tailored learning experiences. EdUJCPS hopes to foster creativity through exploration, scientific inquiry and continuous dialogue, helping students develop AI literacy skills and critical thinking.

Early results show promising outcomes. 65% of students found the maths recommendations from EdUJCPS' chatbot useful, and 60% appreciate the quicker feedback on their homework. Educators have reported that these tools streamline classroom management and identify areas of improvement for more personalised instruction. EdUJCPS plans to expand the use of AI across all grades, building on the early successes.

- How do your current needs align to the driving forces behind EdUJCPS's AI story? What questions does this AI story raise?
- What are the advantages of building your own custom AI applications?
- What training and support might you need to put in place to maximise the impact of AI tools for teaching and learning?

"By adopting a whole-school approach and providing trainings to staff and students, we aim to foster an AI-powered learning setting... AI will take care of the practical tasks... empower[ing] teachers to better meet students' needs, enhancing teaching quality and resulting in a more impactful educational experience."

– Philip K Y Law,
Vice Principal of EdUJCPS

AI Tool: Azure OpenAI Service

Wichita Public Schools

[Visit website](#)

Educators use Microsoft Copilot to make learning more accessible and bring a greater diversity of learning experiences to the classroom.

Increased accessibility

Greater efficiency

With nearly 50,000 students and over 100 different languages spoken, the amount of time and energy required of Wichita educators to individualise their lessons was becoming unsustainable. They needed a solution that could bring diverse, tailored learning experiences into the classroom – swiftly and efficiently.

As existing Microsoft 365 A5 users with Surface devices and Entra ID, the Wichita IT team seamlessly led an early adoption programme of Microsoft Copilot. Educators used generative AI capabilities to increase their efficiency, quickly creating instructional materials that were accessible at different reading levels and in different languages. They also found that they could generate authentic, project-based learning experiences at different levels and streamline individualised student feedback on assignments.

- How do your current needs align to the rationale behind Wichita's story? Is this implementation model a good fit for you?
- What are the advantages of introducing Microsoft Copilot to faculty and staff?
- What AI usage guidelines (privacy, data protection) must be in place before taking the technical steps toward implementation?

"There is a highly documented anxiety 'ping' that affects teachers each Sunday evening. We wonder if we are ready for the coming week and if we have time to get ready. When teachers embrace Microsoft Copilot and begin to understand the time savings it represents, I see the anxiety fade away, replaced by sighs of relief."

– Dyane Smokorowski
Coordinator of Digital Literacy
Wichita Public Schools

AI Tool: Microsoft Copilot

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New York Public Schools

NYC Public Schools

Visit website

A custom AI-powered teaching assistant multiplies teacher effectiveness while reducing burn-out.

Greater efficiency

Insights from data

As the largest public school system in the world, with more than one million students and 1,700 schools, many NYC educators and district staff reported feeling overworked and overwhelmed. The district needed a solution that could help reduce the workload while meeting the individual needs of students and families.

District IT leaders partnered with Microsoft to create a data hub of close to two billion records, forming the foundation for a custom-built AI teaching assistant and family communication tool with Azure AI Foundry. Educators used the AI assistant to scaffold feedback and help students discover answers on their own, multiplying their ability to be several places at once.

- How do your current needs align to the driving forces behind NYC's story? Is this implementation model a good fit?
- What are the advantages of building your own custom AI application?
- What district-level data management solutions must be in place before taking the first steps toward building an AI chatbot?

"Our mission is for students to graduate on a pathway to a rewarding career and long-term economic security, equipped to be a positive force for change. If we are not using AI in education, we're putting our students at risk of being behind."

– Tara Carrozza
NYC Director of Digital Learning Initiatives

AI Tool: Azure OpenAI Service

Department for Education, South Australia

Visit website

Students are supercharging their creativity and critical thinking with AI in the classroom.

Increased accessibility

Greater efficiency

The Department for Education, South Australia is driven by a mission to equip their students for a future where AI is everywhere. Leaders wanted to instil AI literacy and bring generative AI into classrooms, but one question loomed large – how to do it responsibly?

IT leaders relied on Microsoft's Azure AI Content Safety, an AI-powered platform that blocks inappropriate input queries and filters any harmful responses. This allowed them to responsibly deploy EdChat, a custom student-facing chatbot built with Azure AI Studio that is empowering students with the skills they need to thrive in the era of AI. EdChat helps students find quick answers before discussing more complex and nuanced questions with their teachers. Students are also learning how to use AI prompts for feedback on their schoolwork, stimulating their creativity and critical thinking.

- How do your current needs align to the driving forces behind South Australia's AI story? Is this implementation model a good fit?
- What are the advantages of building your own custom AI application?
- Does this model effectively address your stakeholders' biggest concerns when it comes to deploying AI safely and responsibly?

"I think that if we had buried our heads in the sand and banned AI and chatbots in schools, students would likely have continued using it at home to simply generate answers and churn out assignments. By introducing it in schools as part of learning, we're ensuring that they really understand how it can supercharge their thinking and creativity rather than replace it."

– Martin Westwell
Chief Executive of the SA Department for Education

AI Tool: Azure AI Studio

California State University, San Marcos

Visit website

University leaders use Dynamics 365 and the power of AI to establish a personalised connection with every student.

Insights from data

Increased accessibility

As a university with many first-generation students, CSUSM wanted to increase graduation rates and empower social mobility for its diverse population. To do this, they knew they had to find a way to connect with each student, personalise their college experience and meet their individual needs.

CSUSM used Dynamics 365 Customer Insights 'journeys' to tailor the faculty's communications for each student – both digitally and in person – while responding to students' unique interactions and preferences. Dynamics also transformed the school's systems, which were fragmented and siloed, and consolidated their data. University leaders used AI-powered insights to individualise communications and points of interest for every student, resulting in greater attendance and engagement at school-sponsored events and support that continued beyond graduation.

- How do your current needs align to the driving forces behind CSUSM's story? Is this implementation model a good fit?
- What are the advantages of seeking insights into your students' communication preferences?
- Would this model effectively streamline your current data management systems?

"Universities can be complicated for any student, but it can be especially challenging for first-generation students. It's important to know where each of our students are in their lifecycle journeys. To do that, we needed AI technologies that are flexible and can grow with the university."

– Tony Chung
Chief Information Officer
CSUSM

AI Tool: Dynamics 365

Washington State Office of the Superintendent of Public Instruction

Visit website

Leaders take proactive steps toward AI implementation with state-wide guidance and integrated AI teaching and learning standards.

Increased accessibility

Education leaders in Washington state, led by Superintendent Chris Reykdal, are taking proactive steps when it comes to AI use in schools. Washington is among the first states in the U.S. to publish official state-level guidance on AI use in schools, including an implementation roadmap and guidelines for appropriate AI usage for both staff and students.

Driving Washington's AI roadmap is a central human-to-AI-to-human approach: "Start with human inquiry, see what AI produces and always close with human reflection, human edits and human understanding of what was produced." This approach is also helping to drive the development of new teaching and learning standards in ELA, Science and Maths that include AI as an embedded component of the curriculum, rather than being siloed into a separate supplemental area. School leaders are confident that the new standards will provide an opportunity for all students to develop the skills they'll need to be ready for the world of work with AI.

"Our focus remains steadfast on ensuring that every student benefits from these advancements while upholding the highest standards of safety and ethical use."

– Superintendent,
a WA school district

AI Tool: Human-centred AI guidance[1]

[1] ospi.k12.wa.us/sites/default/files/2024-01/human-centered-ai-guidance-k-12-public-schools.pdf

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University of South Florida

Visit website

Faculty and students adopt Copilot for advanced research, data management and administrative efficiency.

Greater efficiency

Increased accessibility

The University of South Florida, a pioneer in research and innovation, has taken a significant step toward ensuring equity. They have become one of the first universities to provide all students, staff and faculty with equal access to technology by fully implementing Microsoft 365 Copilot. This initiative underscores their commitment to creating an inclusive environment where everyone in the organisation has the same opportunities to thrive.

USF adopted a 'platform approach' to AI that includes integrations with all areas of campus life, from student and faculty research to administrative efficiencies. University leaders like CIO Sidney Fernandes have found that, "generative AI and the copilot we have started to use have shown us that a single person can do much more work." With this increased efficiency, faculty and students are finding more time to work on meaningful research, while the IT service desk is focusing more time on solving problems rather than performing triage.

"Copilot now gives me the time that I need that I can be working on building new projects, or really building those relationships. Generative AI is an absolute game changer."

– Tim Henkel,
Assistant Vice Provost/Teaching & Learning Director, USF

AI Tool: Microsoft Copilot

Oregon State University

Visit website

University takes protection to the next level with Microsoft Copilot for Security.

Improved cybersecurity

Greater efficiency

Oregon State University (OSU) is dedicated to conducting open and collaborative research while also prioritising the protection of sensitive data and upholding the institution's reputation. This delicate balance requires a cybersecurity approach that is both robust and responsive.

Partnering with Microsoft, OSU was able to widely implement tools such as Copilot for Security, Microsoft Sentinel and Microsoft Defender quite rapidly. These tools helped the university to use natural language to dialogue across security data to detect and respond to incidents rapidly, reducing response times from weeks to mere minutes. It redefined their approach, shifting from a time-consuming and reactive strategy to a more efficient and proactive one.

- How do your current needs align to the driving forces behind OSU's story?
- What are the advantages of leveraging Copilot for Security to protect your students, staff and their data?
- Would this model effectively streamline your current cybersecurity and data management systems?

"We once had the ability to detect incidents in the timescale of weeks. Now we detect things in matter of minutes."

– David McMorries
Chief Information Security Officer
Oregon State University

AI Tool: Copilot for Security

Sikshana Foundation

Educators leverage generative AI to save time with customised lesson plans.

Visit website

Greater efficiency

Insights from data

India faces challenges such as larger class sizes (average teacher-student ratio of 1:33 versus 1:23 in other countries) and educators managing multiple grades and subjects. The Sikshana Foundation aims to improve education quality by focusing on the concept of 'Shiksha', a Sanskrit term encompassing instruction, lessons, learning and the study of skills.

Understanding the time constraints faced by educators, Microsoft Research India has developed the Shiksha copilot. This mobile-ready tool, powered by generative AI, assists educators in creating personalised learning experiences, assignments and activities.[1] Importantly, it also lightens the workload for educators. The Shiksha copilot, using the Azure OpenAI Service, seamlessly integrates educator insights with curriculum requirements and learning objectives, thereby enhancing efficiency and effectiveness. It is designed to support multiple languages and various input methods, making it accessible to a diverse range of users.[2]

- How do your current needs align to the driving forces behind Shiksha Foundation's story? Is this implementation model a good fit?
- What are the advantages of creating custom copilots to enhance personalisation and alleviate workloads?
- What AI usage guidelines (privacy, data protection) must be in place before taking the technical steps toward implementation?

"Shiksha copilot is very easy to use when compared to other AI we have tried, because it is mapped with our own syllabus and our own curriculum."

– Gireesh K S, Teacher,
Government High School, Jalige

AI Tool: Azure OpenAI Service

[1] timesofindia.indiatimes.com/gadgets-news/microsoft-develops-shiksha-copilot-to-help-indian-teachers-create-study-material/articleshow/104922133.cms
[2] timesnownews.com/technology-science/how-microsofts-shiksha-copilot-is-helping-teachers-in-india-prepare-better-lessons-article-107524567

Indonesia Ministry of Education and Culture

Visit website

Education system uses GitHub Copilot to enhance IT team efficiency and consistency.

Greater efficiency

Indonesia's Ministry of Education, one of the world's largest school systems, serves over 50 million students. With an IT team of only 160 members, or twelve engineers per million students, Indonesia prioritises tools that enhance efficiency and save time on tasks like generating code snippets and creating documentation. GitHub Copilot has enabled the IT team to maintain consistent code and increase productivity without needing to expand the staff.

In 2021, Indonesia launched a Reading Progress pilot programme to combat low literacy rates through personalised feedback and custom passages. Two years later, the Ministry introduced Platform Merdeka Mengajar, utilising Azure OpenAI Service to provide personalised teaching and learning, offering educators high-quality resources and tailored learning paths for students.

- How do your current needs align to the driving forces behind Indonesia's Ministry of Education's story? Is this implementation model a good fit?
- What are the advantages of creating custom copilots to enhance personalisation and alleviate educators' workloads?
- How might your school or institution benefit from improved efficiency and consistency from a tool like GitHub Copilot?

"With just a dozen engineers per million MAUs, maximising the productivity of every engineer is critical for our organisation. We A/B tested the usage of [GitHub] Copilot within our engineering teams, and we found a +42% uplift in development velocity. More than 85% of our engineers also stated that their work is more enjoyable with Copilot's assistance."

– Ibrahim Arief,
CTO of Govtechedu

AI Tool: GitHub Copilot

University of Sydney

Visit website

University developed Cogniti, a secure AI assistant on Microsoft Azure, to enhance student learning safely and boost efficiency.

Greater efficiency

The University of Sydney recognises the importance of generative AI in preparing students for the evolving workforce. They reviewed policies and practices to create clear guidance for appropriate AI use. To address data privacy concerns, they custom-built Cogniti, an AI assistant on the university's secure Azure platform. This ensures prompts and responses remain confidential and are not used for training AI models, safeguarding intellectual property and data privacy.

Developed by educators, Cogniti empowers them to create custom AI chatbots tailored to their instructional needs. The platform enhances student learning through personalised interactions, freeing educators' time for deeper engagement and feedback while also improving prompt writing and AI skills.

The University of Sydney plans to expand Cogniti's capabilities, explore voice interfaces and share the platform with other institutions, setting a new benchmark for AI in education.

- How can involving your educators and staff in tool design, like the University of Sydney's approach with Cogniti, address your institution's needs?
- How might enhancing personalised student interactions and providing deeper learning experiences, similar to the capabilities of Cogniti, address your institution's educational goals?
- How might using an Azure OpenAI tool like Cogniti, help free up educators time to focus on more impactful, personalised student interactions?

"[Faculty aren't] being replaced by technology; their expertise is reflected in the way that it works. Cogniti provides the framework a teacher needs... so that they can strengthen their relationships with students. We want Cogniti to be community developed: built by educators for educators."

– Adam Bridgman
Pro Vice Chancellor of Education Innovation
University of Sydney

AI Tool: Azure OpenAI Service

Section 3

Plan

Provides valuable resources for leaders to prepare AI programs. These resources emphasise proactive planning, covering trustworthy AI frameworks, institutional policies, infrastructure and security and communication strategies. They support initiating conversations, launching planning processes and building community support for successful AI programs.

Introduction to AI

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Leveraging trustworthy AI frameworks

The increasing use and application of generative AI in education holds great promise. Ensuring the responsible use of generative AI in education depends on a combination of the policies, guidelines and frameworks you set and the tools you choose to adopt; in essence, our shared responsibility. For example, safeguarding student data is dependent on both good policies and effective training so educators and students know what to do and how to do it. Leaders should collaborate with educators, policymakers and stakeholders to empower all users with the best that AI technology has to offer while upholding the highest standards of responsibility.

This section of the AI Toolkit highlights the importance of trustworthy AI practices, provides insight into Microsoft's approach to responsible AI and suggests ways to help you get your institution and its policies AI-ready. Questions you may want to consider as you review these frameworks and policy examples include:

1. What goals drive your use of generative AI tools?
2. How does your institution currently manage technology adoption? Will that model work for AI?
3. Should your institution create a single AI policy or adapt existing ones?
4. How will you ensure equitable policy application in AI tool use?
5. What legal considerations must be addressed?

There are a variety of AI frameworks that might help you answer these questions. For example, the resources from Teach AI[1] include sample school guidance materials and frameworks with a focus on reviewing AI output, checking for incorrect information and promoting transparency, safety and respect.

Copilot prompt

Assume the role of an education institution leader such as a provost or superintendent for a medium-sized institution. Provide a list of six policies, frameworks or guidelines (such as Acceptable Use Policies) that should be reviewed and considered for revision to allow for the use of generative AI responsibly and ethically. Additionally, describe three different types of AI use policies that could be developed by schools, universities or ministries of higher education for reference.

[1] teachai.org/

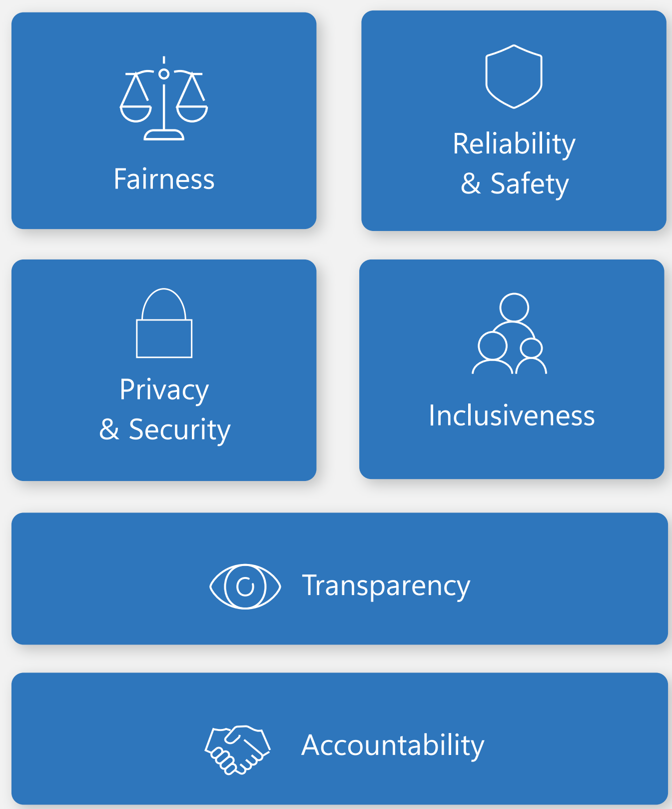
Principles of responsible AI

It is important to consider responsible AI principles when implementing AI in education to ensure these technologies are used responsibly, safely and in ways that enhance educational opportunities for students while preparing them for the future. Microsoft's Responsible AI principles can be used to help inform policies and usage guidelines adopted by districts, states and ministries of education. They can help ensure that AI is used in ways that are fair, transparent and respect the privacy of students and staff.

At the core of Microsoft's AI work are six key principles that guide responsible AI use: fairness, reliability and safety, privacy and security, inclusiveness, transparency and accountability. These principles are embedded in the Microsoft platform, providing built-in tools and frameworks so educational institutions can rely on these values and capabilities as they integrate AI into their operations. For example, Microsoft offers tools for bias detection and data anonymisation.

To successfully implement Responsible AI, both technology providers and users, including leaders, educators and IT professionals, must share responsibility. Microsoft offers tools and platforms with Responsible AI principles in mind, but users must thoughtfully apply these in their specific contexts. This involves regularly reviewing AI applications to protect student privacy, promote fairness and avoid biases while updating policies as AI evolves to maintain community standards.

Microsoft's Responsible AI principles



By leveraging Microsoft's platform and adhering to these responsibilities, educational institutions can foster innovative, effective and responsibly aligned AI environments. Institutions should craft and implement clear policies that guide AI use, establish monitoring processes and ensure all stakeholders understand their roles. These steps help ensure AI supports educational goals, protects rights and privacy and fosters trust and accountability.

Practical steps for education leaders

Translating these guidelines and frameworks into actionable steps is one of the central challenges that educational leaders face when considering generative AI adoption. As you navigate the evolving landscape of generative AI, there are some simple starting points to help you build organisational trust that your AI roll-out will be fair and responsible.

	Suggested actions	Resources
Revise policies to address generative AI	Review policy documentation like Acceptable Use Policies to incorporate language about the use of generative AI.	*Rethinking Acceptable Use Policies in the Age of AI,* *District Administration*
Incorporate AI into teaching and learning	Set guidelines for the responsible use of AI tools in lesson planning and course creation.	*Integrating Generative AI into Higher Education, EDUCAUSE*
Establish monitoring and evaluation standards for AI content	Create a plan to monitor and assess the use of AI.	*ChatGPT and Beyond, Common Sense Education*

Copilot prompt

As a leader in a medium-sized educational institution, such as a provost or superintendent, you are tasked with preparing your institution for the implementation of generative AI. Draft a ten-step plan for integrating generative AI in your educational institution. Focus on policy updates, implementation strategies and evaluation methods to ensure a smooth transition.

Engaging the community

AI technologies are revolutionising education, presenting exciting opportunities for teaching and learning. However, implementing AI-driven tools requires thoughtful planning, clear communication and collaboration with supportive communities. Stakeholders – including administrators, educators, students, parents and community members – have diverse responsibilities and seek understanding of how AI tools will influence their daily experiences and future endeavours. This section addresses significant challenges and opportunities in involving your community in the adoption of AI-powered education tools. It also offers practical advice and strategies to:

- Build trust and support for AI-powered tools in education.
- Understand and address your community's concerns.
- Match tools to your goals and needs.
- Build a shared vision with your community.

Building trust and support with stakeholders

Building trust with stakeholders across educational communities is a crucial aspect of any initiative. There are many ways to build this trust.

- Seek feedback and input from diverse stakeholder groups.
- Utilise appropriate translation services as required for speakers of other languages.
- Align with shared objectives and values that prioritise student success by offering high-quality learning opportunities.

Familiarising yourself with the key points below will empower you to engage in meaningful discussions with your community partners.

Key point

AI-powered tools help faculty and staff automate or streamline time-consuming tasks that interfere with more crucial learning-related needs.

Responding to emails, exploring data trends, researching new instructional approaches and drafting detailed syllabi and course information take time away from connecting with and meeting diverse students' needs. Generative AI tools give educators time back so that they can refocus on what matters most. Learn how educators in Wichita Public Schools used Copilot to become more efficient in the *AI Navigators tab.*

Key point

AI-powered tools can also help address some of the critical challenges and opportunities in education.

Accessibility is a key component of equitable schools. Generative AI tools can help educators create high-interest text for emerging readers, develop multiple means of representation for content and offer new ways of demonstrating ideas for students. This creates more engaging and equitable learning experiences for students. Read about how educators in Wichita Public Schools used Copilot to adapt primary documents for their social studies classes in the AI Navigators tab.

Transformative workplace skills

Understanding how AI is already impacting the workplace, classrooms and lecture halls is critical to preparing students and your community for adopting AI. As a fundamental component of the fourth industrial revolution, AI – along with related fields such as machine learning and data analytics – is transforming workplace skills and experiences.[1] Medical research centres, various businesses, municipal operations and sustainable energy science are all driving rapid innovation. For instance, Walmart uses AI technologies to streamline inventory management, ensuring the availability of the correct products for their customers.[2] Walmart also partnered with nearby Bentonville Schools to provide AI learning experiences for local students.[3]

To address evolving workplace needs, many schools and institutions have implemented a multi-tiered approach. This includes the recent introduction of a K-12 vertical programme that integrates AI principles into every grade level and subject area.[4] Across campuses nationwide, billions of dollars have been allocated to develop programmes, recruit faculty and construct buildings to establish new AI initiatives aimed at driving ongoing innovation.[5]

For instance, in early 2023, the University of Buffalo launched the National AI Institute for Exceptional Education.[6] Their initial projects include the AI Screener, which identifies each student's needs and the AI Orchestrator, which assists speech and language pathologists in creating personalised interventions.

[1] mckinsey.com/featured-insights/mckinsey-explainers/what-are-industry-4-0-the-fourth-industrial-revolution-and-4ir

[2] https://tech.walmart.com/content/walmart-global-tech/en_us/blog/post/walmarts-ai-powered-inventory-system-brightens-the-holidays.html

[3] core-docs.s3.amazonaws.com/documents/asset/uploaded_file/3173/Bentonville/2197839/News_Release_Ignite_Program_Expansion.pdf

[4] schools.gcpsk12.org/page/32147

[5] insidehighered.com/news/tech-innovation/artificial-intelligence/2023/05/19/colleges-race-hire-and-build-amid-ai-gold

[6] buffalo.edu/ai4exceptionaled.html

Understanding and addressing your community's concerns

As you meet with different community members, you'll encounter a multitude of concerns, interests and needs. Use this opportunity to build empathy by actively listening to their questions, demonstrating your AI expertise and leadership and inspiring their support for your AI initiative.

Leadership and administrators

Schools are increasingly the target of cyberattacks aimed at accessing student data. Consequently, school leaders are dedicating more thought, resources and funding to protecting user data.

"Student privacy is one of our biggest concerns especially when it comes to AI tools. We are committed to vetting any tool before it's introduced to our classrooms to make sure that we know how our data is used and that, ultimately, it's safe and protected. We're able to use Copilot for Security to enhance and extend our ability to identify threats, automate our responses and remediate any issues."

School leaders may have concerns about equity and accessibility when it comes to integrating AI. Schools want to ensure that AI tools are accessible to all students, including those with disabilities, and that these tools do not worsen existing inequalities.

"We will evaluate all AI tools to make sure they can be equitably accessed and used by students from various socio-economic backgrounds, different levels of technology access and diverse learning needs. We aim to understand how AIcan help us build a fairer educational landscape and remediate any issues, as exemplified by institutions like the University of Texas."

Educators and practitioners

Based on past experiences, educators may feel that new programmes and initiatives are introduced, supported briefly and then forgotten. Some teachers hesitate to adopt new technology unless they feel confident with their own skills and can address any questions or issues their students might encounter.

"We are committed to making sure that you and your students know how to use AI tools responsibly. Our plan includes age-appropriate instructional materials, suggested conversation starters, guidance on modelling appropriate use and taking an iterative approach to adapting policies. You can also refer to resources like Microsoft Learn's Equip your students with AI and tech skills for today – and tomorrow and Empower educators to explore the potential of artificial intelligence courses for self-paced learning."

Educators focus on the positive impact of instructional strategies and tools on learning. They are generally more receptive to concepts and tools that are easy to adopt and show both immediate and lasting effects.

"Early research indicates that students benefit from AI-generated explanations, outperforming those who only receive correct answers.[7] We also have Learning Accelerators available that provide learners immediate, personalised coaching to help develop foundational and workplace skills."

[7] papers.ssrn.com/sol3/papers.cfm?abstract_id=4641653

Students and families

Families may have reservations about large corporations profiting from children's data and AI queries. Remembering past instances of broken trust, they want to shield their children from similar experiences.

"We prioritise your student's privacy by thoroughly examining each company's privacy policies. We ensure that their data practices align with our values, prioritising options that prioritise privacy and responsible data use. We believe privacy is a fundamental right and only work with providers who share this belief."

Families rely on schools to equip their children for future aspirations and careers. They expect students to have access to the latest technologies and opportunities that pave the way for a successful transition into adulthood.

"We've integrated AI features into the tools students use daily for learning, creativity and productivity. Additionally, we're exploring how other schools have implemented AI guardrails. These guardrails help students access school-specific chatbots designed to support their individual learning requirements."

"Integrating AI tools into our instruction is part of our commitment to preparing students for the future. Experts at the World Economic Forum and McKinsey & Company have highlighted AI's significance in defining the workplace.[8,9] By incorporating AI into our teaching methods, we're ensuring that our students have the skills they need to thrive in this evolving landscape."

[8] weforum.org/press/2020/01/the-reskilling-revolution-better-skills-better-jobs-better-education-for-a-billion-people-by-2030
[9] mckinsey.com/featured-insights/mckinsey-explainers/what-are-industry-4-0-the-fourth-industrial-revolution-and-4ir

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The wider community

In addition to maintaining a high-performing school system and ensuring student safety, the broader community expects their tax dollars to be used efficiently and responsibly.

"We've been looking into how AI can enhance our data analysis efforts in schools. What we've discovered is quite promising. With AI-powered tools, we can analyse data in ways previously unimaginable. This means we can make informed decisions on how to optimise our resources and staffing to best support our students. Whether it's adjusting bus routes, optimising utilities or refining staffing allocations, AI enables us to pinpoint areas for improvement and redirect our focus and funds toward what truly matters: the learning experience of our students."

AI tools have limitations and sometimes produce inaccurate information. We want our students to graduate with solid knowledge and useful skills, so we should be careful about how we use AI in our classrooms, if at all.

"We're excited to announce our plans to introduce age-appropriate, custom chatbots for our students, inspired by successful initiatives like those in New York City Public Schools. These chatbots will be tailored specifically for our students, ensuring that the data used by the model comes from trusted sources and that we have control over our users' data. This means our students will have access to AI tools in a protected and safer environment compared to public tools. We're committed to providing innovative yet secure learning experiences for our students."

Continuing the conversation

No matter where you are in the process, you'll continue to speak with a wide variety of stakeholders – including educators, instructional leaders, school or college administrators, students, families and community members. Maintain strong relationships with these thought partners with open dialogue.

How can I protect the privacy and security of students' data when using AI-powered tools?

"We start by reviewing each AI tool's terms of service and privacy policy to ensure that they are committed to privacy and are aligned to our expectations. We know that Microsoft's generative AI solutions like Copilot, Microsoft 365 Copilot and Azure AI Foundry support FERPA compliance and student data privacy protection. They use advanced encryption and data handling policies to secure sensitive information. Microsoft's AI solutions provide access controls and transparency in data usage, undergoing regular compliance audits to maintain high standards of privacy and security. We can customise privacy settings to align with our specific compliance requirements and data governance policies."

How do these AI solutions enhance student engagement and improve learning outcomes? Are there studies or evidence demonstrating their effectiveness in educational settings?

"AI's impact on instruction will keep evolving as practices, tools and usage grow. Early findings show that including generative AI explanations of concepts in instruction has a positive effect on learning when compared with just giving corrective feedback."[10]

How can I prevent academic dishonesty and plagiarism when using AI-powered tools?

"Protecting our school's academic integrity begins with all users learning how to use AI responsibly. We're starting with professional development for educators, modelling responsible use and having open discussions. We've also paired our training with a clear policy."

How can these AI solutions be customised to align with our district's specific curriculum standards, instructional goals and educational objectives?

"We're able to create customised AI-powered teaching assistants and configure data dashboards that help address our unique challenges. Even better, the data that we use in our schools is kept private within the organisation and is not used to train larger models."

How do these AI solutions accommodate the language differences among students, educators and faculty in our institution?

"Microsoft's AI solutions, such as Copilot and Azure AI Foundry, are designed to support multilingual environments. They offer features like real-time translation and multilingual support across various applications in more than 100 languages. This ensures that students, educators and faculty can engage with content in their preferred language, enhancing comprehension and participation. Additionally, these tools allow for the customisation of language settings, enabling institutions to align AI-powered resources with the diverse linguistic needs of their communities."

[10] papers.ssrn.com/sol3/papers.cfm?abstract_id=4641653

How can I support students with diverse learning needs and preferences when using AI-powered tools?

"Educators can use tools like Copilot and the Learning Accelerators to personalise instructional content to meet individual student needs. With Copilot, educators can quickly adapt content into different languages or reading levels. Furthermore, they can use prompts to create custom explanations or analogies that build upon age-appropriate knowledge or a student's interests. Copilot supports multiple means of generating prompts including text- or voice-based and Microsoft 365 Copilot includes screen reading capabilities."

Learn more at the Accessibility tools for Microsoft 365 Copilot resource page.

Your empathy, teaching expertise and strong communication skills will support the successful implementation of AI-powered tools in your school or institution. Continue collaborating with stakeholders and practising transparency to address your community's concerns, needs and goals.

Institutional policy considerations

Practical preparation

Establishing policies creates structure and guidelines for your faculty, staff, students and community. Before you get started, consider these practical suggestions.

- Start now. Your students and staff are likely using AI already and need guidance. Create initial policies and iterate as you go.
- Identify key areas of need and critical questions that will guide your process.
- Establish what needs policy and what doesn't. Focus on the largest areas of impact.
- Learn from peers and familiarise yourself with resources like the TeachAI toolkit, developed with support from Microsoft.[1]

As your school or institution develops its AI strategy, it's natural to shift your focus affected areas, especially policies that may need updating to address recent changes. Start by consulting government guidelines and requirements and reviewing your existing policies. Then, consider curating a set of exemplary policies that can be customised to meet your specific needs.

[1] teachai.org/toolkit

Organisation policy considerations

Crafting, updating and approving new policies is a critical task. We encourage you to take an iterative approach to policy development, emphasising ongoing improvement. In the case of AI, schools are still exploring the full extent of its use and impact, and a successful policy is one that is regularly reviewed and revised to meet the current needs of the school and community.

This section begins by examining a hypothetical academic integrity policy and outlines how a school could enhance it for better guidance on students' use of AI. The latter part offers concise recommendations for reviewing other policies.

Academic integrity policy

Early research reveals that students who have access to explanations created by generative AI perform better than students who only have access to correct answers.[2] AI's impact goes beyond tool usage and user considerations. It involves how every member of the school or higher educational community adopts and uses AI tools. Your school, like many others, aims to define clear guidelines for academic integrity considering increased student use of generative AI tools. It's crucial to evaluate the effects of policies, whether highly restrictive, fully encouraging or a mix of both, on students, their education and the overall fulfilment of academic standards.

Questions to lead your discussion

- Are your students allowed to use AI on assignments?
- Which policy model will guide your practice?[3]
- What impact will that have on your current academic integrity policy?

Policy spotlight

South Australia's Department for Education led a pilot programme that introduced a custom chatbot for students to use. Their policy provides structure and guidance around how learners can responsively use generated content.[4]

[2] papers.ssrn.com/sol3/papers.cfm?abstract_id=4641653
[3] https://www.education.sa.gov.au/parents-and-families/curriculum-and-learning/ai
[4] education.sa.gov.au/parents-and-families/curriculum-and-learning/ai#:~:text=AI%20and%20AI%2Denabled%20technology,security%20and%20students'%20learning%20needs.

Academic integrity highlight: Refining a policy

Here is one possible evolution of an academic integrity policy. Each step in the evolution includes a sample policy which is followed by a quick analysis of its effectiveness.

Initial policy

Presenting another person's work as your own is an act of dishonesty. This behaviour undermines your integrity and contradicts the principles upheld by [our institution]. We maintain the belief that academic success is contingent upon the dedication you invest in your studies.

Analysis

This policy addresses human-authored texts. Given the many ways that students can use generative AI tools, clear guidance on responsible AI use is essential to maintain academic integrity and prevent plagiarism.

Revised policy

Presenting another person's work or content created by a generative AI tool as your own is an act of dishonesty. This behaviour undermines your integrity and contradicts the principles upheld by [our school]. We maintain the belief that academic success is contingent upon the dedication you invest in your studies. We expect you will approach your assignments honestly, as your work reflects your capabilities.

Analysis

This policy covers generative AI. It broadens the range of permitted uses for students beyond mere assignment copying, but does not offer appropriate uses for AI. We recommend setting guidelines for additional generative AI uses like revising work, seeking formative feedback and utilising AI as a brainstorming partner.

Leadership teams can create prompts to assess existing policies for improvement and explore various wording options. For instance, Copilot can analyse a revised policy, review it for potential biases and request a simplified version in plain language accessible to all student and community groups.

Apply your learning

Open your institution's academic integrity policy in the Edge browser. Open Copilot sidebar from the top right and enter the prompt below:

As the CAO of a school district, analyse our existing academic integrity policy, focusing on AI's ethical use by students. Evaluate the policy's current consideration of implicit biases, linguistic, cultural and socio-economic diversity. Suggest concrete, actionable improvements to enhance inclusivity, fairness and clarity, ensuring the policy is understandable and accessible to all students. Provide examples of best practices from other policies and include a revised policy draft incorporating these elements.

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Sample policy developed with Copilot

At [our school], we prioritise academic integrity as a core principle. We anticipate that all students will complete their assignments with honesty, understanding that their work showcases their abilities. Additionally, we emphasise responsible AI usage, which involves thoroughly reviewing content rather than merely copying and pasting generated material. To promote responsible AI use and ensure fairness, we provide the following guidelines.

Attribution and AI content

- When using AI-generated content, always provide proper attribution.
- Presenting AI-generated work as your own is strictly prohibited. Faculty will set clear expectations regarding responsible AI use for their class using approved categories. Sample policy language:
  - ✓ Highly restrictive: No AI use is allowed.
  - ✓ Fully encouraging: AI is fully available for student academic use. No restrictions.
  - ✓ Hybrid: AI use is for brainstorming and AI tutorials, but not for submitted assignments.

For additional information or language, refer to the AI guidance & FAQs from Harvard.[5]

Cultural sensitivity

- We respect diverse cultural norms related to collaboration and attribution.
- Students should be aware of these differences and adapt their practices accordingly.

Implicit bias awareness

- Our academic integrity process aims to be unbiased and consider individual circumstances.

Education and resources

- We offer workshops, tutorials and online resources on citation practices and responsible AI use.
- Students are encouraged to learn and apply these principles.

Equitable enforcement

- Violations will be addressed consistently, regardless of socio-economic status or cultural background.
- Fairness and equity guide our approach.
- If assignments allow or require AI use or specific tools, those tools must be readily available and provided for each student.

By adhering to these guidelines, students contribute to a respectful and inclusive academic community. Let's work together to foster integrity, learning and growth.

Analysis

This policy outlines expectations for students, emphasising the many ways that students may use generative AI tools in their workflow. It reflects the school's dedication to fairness and outlines efforts to teach students responsible AI skills.

[5] oue.fas.harvard.edu/ai-guidance#:~:text=POLICIES%20FOR%20THE%20USE%20OF%20AI%20IN%20COURSES

Additional policy considerations

After exploring the process of evaluating and drafting an academic integrity policy, apply similar methods to other policy areas. Consider how your team can implement the methods discussed earlier in these new contexts.

Data protection and privacy

Large language models (LLMs) depend on user data to produce results. It's important that data and privacy are core considerations when approving AI tools. Further, take the time to transparently communicate to students, staff and the community about how the school and tools use and protect their data. Use these guiding questions to lead your discussion:

- What does student privacy mean in the AI era?
- How well do our data protection and privacy policies align with legal regulations?
- How do we communicate our data usage policies and practices to students, staff and families? Is there an opt-out option?

Staff and faculty use

AI tools can enhance educator efficiency and personalise student content. Clear guidelines for AI tool usage are highly recommended. The benefits of generative AI for educators range from creating lesson plans, curating content, automating tasks and generating communication (e.g., emails). Use these guiding questions to lead your discussion:

- How might we improve learning by using AI for instructional purposes?
- What instructional uses do we want to encourage? What might we restrict?
- How will we support our staff through professional learning?

Policy spotlight

Educators in Wichita Public Schools have used Copilot to develop instructional resources and individualise their students' learning. In this scenario, the educators would benefit from a combination of focused professional learning alongside established guidelines in the district's policies.

Classroom syllabi

Educators use syllabi to communicate expectations, instructional resources and experiences, often incorporating academic integrity statements. Consider offering educators a standardised statement for inclusion or adaptation in their syllabi. Use these guiding questions to lead your discussion:

- What message should be included on all class syllabi?
- How can this statement reinforce broader policies?
- To what extent can educators adapt the statement for their classes?

Accessibility and Universal Design for Learning (UDL)

AI tools have the potential to make learning more accessible for all learners. From simply adapting content into more accessible formats to creating instructional materials tailored to individual student's needs, AI tools offer great promise for making learning more accessible and equitable. Use these guiding questions to lead your discussion:

- What are the accessibility and language proficiency needs of our students and staff?
- How might AI tools help us enhance accessibility for all learners
- What government guidelines must we follow as we evaluate AI tools and design our school's AI program?

Policy spotlight

California State University San Marcos expanded the traditional use of 'accessibility' by creating AI tools that enabled custom communication to better reach diverse groups of students such as first-generation college learners.

The importance of a secure foundation

Education leaders know that protecting data and preventing cyberattacks are essential for safe, secure and effective learning environments. Data-rich organisations like schools, universities and ministries of education are increasingly targeted by cybercriminals, as evidenced by rising attacks and evolving social engineering threats.

According to Microsoft Cyber Signals report:

- Education is the third most targeted industry
- Educational institutions face an average of 2,507 cyberattacks per week
- Over the past year, Microsoft Defender for Office 365 blocked more than 15,000 emails per day targeting the education sector with malicious QR codes – including phishing, spam and malware.

The U.S. Cybersecurity and Infrastructure Security Agency (CISA) has launched a campaign to address the spike in cyberthreats impacting education. Beginning with the *Protecting Our Future* report, federal security experts and the Department of Education outlined key recommendations that increase security and privacy in schools.[1] In late 2023, President Biden codified additional ways AI systems must be secured when he signed the Executive Order on the *Safe, Secure and Trustworthy Development and Use of Artificial Intelligence*.[2] The Executive Order called for public and private organisations to make sure that AI systems are resilient against misuse and modifications, function as intended and are ethically developed and deployed in a secure manner.

Schools and universities across the country are listening to the government's calls for increased cybersecurity measures and closer examination of AI systems' security and privacy. Many states are adopting policies for safe AI use in K-12 school districts, with help from companies like Microsoft and government agencies. Microsoft is also working closely with higher education institutions like the University of Michigan to deploy copilots that are as secure as other existing infrastructure.[4]

[1] cisa.gov/resources-tools/resources/report-partnering-safeguard-k-12-organizations-cybersecurity-threats

[2] whitehouse.gov/briefing-room/presidential-actions/2023/10/30/executive-order-on-the-safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence/

[3] er.educause.edu/articles/2024/2/how-and-why-the-university-of-michigan-built-its-own-closed-generative-ai-tools

[4] doi.org/10.1016/j.platter.2023.100779

With the 2023 launch of the Secure Future Initiative (SFI) Microsoft accelerated its development of AI-based cyber defence to provide machine-speed security for educational institutions. But what can you do if you want to provide secure AI systems, but don't know where to start?

This section of the AI Toolkit offers suggested actions to help you implement generative AI tools safely and securely. You'll also discover how Microsoft's AI systems and A3/A5 Microsoft 365 Education plans can enhance your security with tools designed to give you the necessary control and protection to effectively manage AI within your school's infrastructure.

Understand the importance of a responsible AI framework

A secure AI foundation begins with reviewing responsible AI practices and understanding any potential issues with using AI in your educational setting. As you evaluate AI systems, consider the following questions:

1. Does the AI system follow responsible AI practices?
2. How did developers and designers follow responsible AI practices when developing their solutions?
3. Does the AI system align with jurisdictional laws and regulations for my school or institution?

Responsible AI implications

It's important to research and understand how generative AI systems are built before developing or deploying a tool in your environment. Consider this to ensure an example from K-12 education.

> Stanford researchers found that seven different GPT systems falsely identified over 50% of Chinese students' essays for an 8th grade Test of English as a Foreign Language (TOEFL) as 'AI-generated' when they were not.[5] There were no inaccuracies when the GPT systems analysed 8th grade essays written by native English speakers.

Fairness issues – like assessing writing differently when a learner is a non-native speaker – is a responsible AI principle that should be examined when considering AI systems. Suppliers should be able to explain how their systems address responsible AI. Consider asking questions about their development process or the steps they take to ensure their AI system is responsibly designed.

Microsoft develops AI systems using Responsible AI principles. From development to deployment and beyond, engineers and technicians follow the Microsoft Responsible AI Standard to check that AI systems provide similar quality of service for identified demographic groups, including marginalised groups.

Before moving on, evaluate how AI systems generate content, and review available research on potential bias in outputs.

[5] https://hai.stanford.edu/news/ai-detectors-biased-against-non-native-english-writers

Identify outcomes and data sources for AI systems

It can be helpful to create a list of the desired outcomes you want the AI system to accomplish and what data might be required. Keep the following questions in mind.

1. What are some pain points in your school or institution?
2. What needs do community members have that an AI system might address?
3. How does the AI system use data to generate responses?

Goals

AI systems perform different functions and have different capabilities. For example, Microsoft Copilot helps people increase their productivity and creativity, whereas Copilot for Security provides critical incident response, threat hunting, intelligence gathering and posture management for IT administrators. Knowing what you want the AI system to accomplish, your goals, will help you find the right solutions for your institution.

It's helpful to begin by making a hierarchical list of needs or pain points that an AI system might address. Place the most urgent items at the top of the list. You might want to ask colleagues from other departments to offer input on what you identified.

After you have your list, consider rewriting the pain points into goal statements. For example:

Pain point: IT administrators struggle to prioritise threats because of the number of signals that emerge each day.

Goal statement: Any security-focused AI system should help administrators prioritise threats and give guidance on steps to take to respond appropriately.

Compare your goal statements against the capabilities of any AI systems that you are considering. If there is alignment, evaluate how the system meets trustworthy AI requirements to ensure it aligns with your institution's values and needs.

Data sources

AI systems use data to generate responses. Sometimes, AI needs access to files on your computer or network, like when you ask Microsoft 365 Copilot to summarise important points from a notes document. In other cases, AI requires access to critical systems to perform specialised tasks. For example, when building a custom copilot with Azure AI Foundry, you have options to connect private data sources to increase customisation and deliver fine-tuned responses. Regardless of which AI system meets your goals, be mindful of the data requirements and laws that govern data privacy in your jurisdiction.

The following activities can help you make informed decisions about data handling.

- Form a committee that includes compliance officers, security administrators and other leaders. Review and revise the goals you wrote, identify any data sources required in the AI systems under consideration and list any compliance requirements that must be met before implementation.
- Draft a list of data governance questions that you can ask suppliers. Key topics include what data sources are required, how data is kept safe and what mechanisms help you manage risk.
- Review Microsoft's enterprise data protection for Microsoft Copilot and Microsoft 365 Copilot. With Microsoft's Enterprise Data Protection (EDP), both prompts and responses are safeguarded by the same contractual terms and security commitments that customers have long trusted for protecting their emails in Exchange and files in SharePoint.

Before moving on:

✓ Align your stated goals with the capabilities of AI systems under consideration.

✓ Determine what data is required to use an AI system.

✓ Evaluate data requirements against any laws or regulations that govern your school or institution.

Establish data governance, roles and responsibilities

After you have identified an AI system that's secure, compliant and addresses a goal, begin exploring data governance for your school or institution. Keep the following questions in mind.

1. Does my school or institution have the infrastructure required for AI applications to access data securely, quickly and at scale?
2. What infrastructure and resources are available to support AI deployment?
3. Who is going to be responsible for ongoing monitoring, troubleshooting and communication?

Develop a plan and consider roles

It's important to assess your infrastructure's readiness for safe, secure AI experiences. Sometimes you will find an AI system that meets or exceeds the security requirements dictated by laws, but your infrastructure or staff are not prepared to safeguard and monitor interactions. Having a plan will help you think through data governance issues that might be specific to your school or institution.

Here are some questions to consider as you develop your plan.

- Would it be better to buy a pre-built AI system, develop AI applications in-house or update existing AI systems?
- Should data for AI systems be stored on-premises or in the cloud?
- What data platform architecture is necessary to run the AI system? Do the requirements match what is allowed by law for my school or institution?

In addition to assessing infrastructure capabilities, you also need to examine the extent to which your IT administrators can monitor and assess any AI system that goes into your environment. The Cyber Security and Infrastructure Security Agency (CISA) recommends establishing an incident manager, technology manager and a communication manager to oversee AI systems.[6]

[6] https://www.cisa.gov/sites/default/files/publications/Incident-Response-Plan-Basics_508c.pdf

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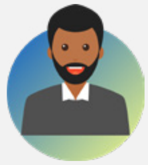
The **incident manager** leads the response when an incident with an AI system occurs. They manage communication flows and delegate tasks, but do not perform any technical duties.



The **technology manager** is the subject matter expert for all AI systems. They should have knowledge of AI, data security and appropriate response measures.

The **communication manager** communicates with internal and external stakeholders about important decisions or incidents.

If you can't identify who might serve in these capacities, consider hiring someone who might have the expertise to perform the functions. These roles are important! Assigning people to roles and including them in any implementation plans will help you gather diverse perspectives and build teamwork well before the school community begins using the AI system.

The Microsoft 365 Education A3 and A5 plans and security add-ons include applications that help monitor AI activities and data flow. For example:

- Microsoft Defender for Cloud: Monitor AI system usage across cloud, multi-cloud or hybrid infrastructures, understand associated risks and approve or block access by browsing a catalogue of 400+ generative AI applications.
- Microsoft Purview: Detect data security risks in Microsoft 365 Copilot through Purview's AI hub. The AI hub aggregates usage statistics and applies a risk level to over 100 of the most common AI applications. Purview also uses sensitivity label citation and inheritance for additional security with AI systems.
- Microsoft Purview eDiscovery: Identify, preserve and collect relevant AI data and interactions litigation, investigations, audits and inquiries.

Before moving on:

✓ Evaluate your organisation's Zero Trust security posture and overall capabilities.

✓ Determine data management requirements like cloud-based or on-premises storage.

✓ Identify colleagues who have the expertise to serve in AI-specific roles.

Determine data privacy procedures and safeguards

Data privacy is a central concern when using AI systems or any technology in K-20 settings. Understanding how data is kept safe and secure is one piece of information that you need to assess. It's also important to review internal policies and identity access protocols prior to deploying an AI system. Keep the following questions in mind.

1. What are the known privacy risks with the AI system?
2. How is data shared, used and stored in the AI system?
3. How do people access and use the AI system?

Privacy impact assessments

Privacy impact assessments (PIA) are tools that government agencies use to evaluate information technology (IT) systems for privacy risks and identify options for mitigating these risks. Learning how agencies approach evaluating privacy and data risks in a PIA can help you assess solutions you are considering. A privacy impact assessment typically includes:

- Known privacy risks.
- Options for mitigating known privacy risks.
- Instructions on how to properly handle privacy issues.
- Processes for analysing the legal compliance with privacy laws and regulations.
- Documentation on the flow of personal information.
- Public assurances that personal information is protected.

If you can address each one of these points when you evaluate an AI system, you'll have enough information to make an knowledgeable decision about data privacy protection.

Privacy policies

Vendors should clearly articulate how data is used, stored and shared when discussing AI solutions with you. After all, schools and institutions are responsible for helping protect student privacy, so AI solutions must provide adequate security and protection built into the application design.

It's wise to ask suppliers for data privacy documentation or a legal Privacy Policy for any AI systems you are considering. Reviewing data privacy and security statements can help you make informed, legal decisions about what's best for your school or institution. For example, Microsoft publishes how data is used in each one of its AI systems:

Microsoft Copilot

Copilot experiences in Windows

Microsoft 365 Copilot

Copilot for Security

Azure OpenAI Service

Infrastructure settings

Your computing infrastructure and security services can add additional layers of privacy protection, depending on how they're configured. Offering secure identity access protocols and creating user polices are two ways to increase privacy without relying solely on built-in protections of AI systems. Take time to speak with IT administrators to understand your system's capabilities.

Microsoft offers two solutions that help you maintain data privacy when using AI systems:

- Microsoft Entra ID: Manage access to Microsoft Copilot and underlying data with secure authentication procedures and risk-based adaptive policies.
- Intune for Education: Apply security, configuration and compliance policies to devices so that school-issued endpoints have baseline protection when working with AI systems.

Consider asking suppliers how their AI systems integrate with Microsoft solutions or other products you use to control access and manage policies.

Before moving on:

- Use the parts of a privacy impact assessment to assess your institution's data privacy protocols.
- Obtain a copy of any privacy and security documentation for the AI system.
- Evaluate your existing infrastructure to strengthen identity access control and implement security policies.

Develop an incident response plan to address issues that arise

Having an incident response plan ensures that you can respond effectively when an issue arises. Even the most secure infrastructure can experience incidents, so having a response plan before you launch an AI system will help you think through necessary logistics and procedures ahead of time. Keep the following questions in mind.

1. What constitutes an incident with an AI system?
2. What parts go into an incident response plan?
3. Who should be notified when an incident occurs?

Incident response plans

Before diving into incident response plans, you should understand what constitutes an incident. Microsoft defines an incident as a group of correlated alerts that humans or automation tools deem to be a genuine threat. Although one alert on their own might not constitute a major threat, the combination of alerts might indicate a possible breach.

Well-designed, secure AI systems that run inside carefully managed infrastructure still face threats that lead to incidents. Some common points of failure include:

✓ Security breaches that give cyber criminals access to sensitive data.

✓ Unintentionally disclosing private information that would not otherwise be shared.

✓ Discriminatory or misleading information in responses.

Developing an incident response plan that handles issues like these will enable you effectively address issues that arise. CISA recommends schools and institutions layout a six-stage incident response plan.

1. **Preparation**

   Document and share policies and procedures for incident response, configure security systems to detect suspicious and malicious activity, create roles and assign responsibilities and educate users about the AI system.

2. **Detection and analysis**

   Define implementable processes, gather baseline information to monitor and detect anomalous or suspicious activity and outline the differences between an authorised use and an incident.

3. **Containment**

   Develop an approach to contain or minimise threats, and identify known security containment strategies for common incidents.

4. **Eradication and recovery**

   Outline how to eliminate artefacts from an incident, create mitigation steps to address exploited vulnerabilities, document evidence collection procedures, establish a regular back-up plan and list steps for recovery and return to normal operation.

5. **Post-incident activity**

   Outline how incidents should be documented and reported, identify steps for hardening security and develop ways to share and apply lessons learned.

6. **Coordination**

   Identify who needs to be informed when an incident occurs, depending on the severity of the threat.

A great way to create an incident response plan is to form a committee with colleagues who have expertise in each area. If you identified an incident manager, technology manager or communication manager, invite these people to contribute to the incident response plan for AI systems.

For additional information on developing your own cybersecurity incident response plan, check out CISA's Incident Response Plan (IRP) Basics or the K-12 Six Essential Cyber Incident Response Runbook.[7,8]

[7] cisa.gov/sites/default/files/publications/Incident-Response-Plan-Basics_508c.pdf
[8] static1.squarespace.com/static/5e441b46adfb340b05008fe7/t/62cc8e3843251c6d2b2cb0a5/1657572921164/K12SIX-IncidentResponseRunbook.pdf

Data governance

Ensuring the security and integrity of your data assets is a top priority for organisations across all sectors. Building a strong security posture includes a well-defined data governance framework. Data governance and security are fundamentally intertwined, each reinforcing the other to safeguard the confidentiality, availability and integrity of data. By combining effective data governance with robust security measures, organisations can defend against a wide range of cyber threats, ensuring that their data is both well-managed and highly secure.

Cloud data consolidation

In an educational setting, migrating data to cloud storage offers significant advantages. A cloud data governance framework acts as a strategic blueprint, guiding the storage and management of your data in the cloud. Tools enable your team to monitor and understand data movements, while your governance framework establishes the rules, roles, procedures and processes for securely managing and controlling these data flows within cloud storage. This approach helps ensure safe and efficient data handling in schools.

Data governance and privacy needs

As Information and Instructional Technology leaders, you face a delicate balance between needing to manage data safety and privacy while adopting new technology. Your expertise is pivotal in creating a secure, effective and innovative educational environment. This includes responsibly adding AI into your schools, ensuring it not only improves learning, but also respects and safeguards the privacy and integrity of all data and individuals involved.

While generative AI has many potential applications in education, it also poses significant challenges for data governance and privacy, as it may involve processing sensitive or personal data. This document provides a practical guide for you on addressing these challenges and implementing generative AI responsibly.

Data governance in AI

Data governance involves defining and implementing policies, standards and practices for managing data quality, security and compliance. This also helps establish processes that keep your data secured, private, accurate and usable throughout its life cycle.

A strong data governance strategy is essential for any organisation that relies on data to improve decision-making and ensure successful outcomes. When collecting vast amounts of data, a strategy should be in place to manage risks, reduce costs and effectively support organisational objectives. Data governance is essential for ensuring that generative AI is used for legitimate and beneficial purposes, and that the risks of data misuse, leakage or corruption are minimised.

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In the context of AI, data governance transcends regulatory compliance; it forms the foundation for responsible and effective AI deployment in education. Evaluate your existing framework with these key questions in mind.

- **Data security:** Can you detail your organisation's current data security measures, including documentation processes and data loss prevention efforts?
- **Data policies:** Have you developed and implemented comprehensive data policies to govern the use and sharing of data within your AI systems? How do these policies ensure transparency and uphold responsible standards, especially when handling sensitive student data?
- **Data sources:** Have you reflected on the diversity and reliability of the data sources underpinning your data governance framework? How do these sources impact the integrity and effectiveness of your data management strategies?
- **Data compliance:** How does data compliance, beyond mere adherence to regulations, play a foundational role in your data governance framework? How might this influence the efficacy and responsibility of your data management practices?
- **Data quality:** What measures you've taken to ensure the accuracy, consistency and reliability of data? How do you verify the integrity of student performance data, demographic information and the relevance of learning materials?
- **Data management:** Reflect on your data management practices, specifically regarding the handling, storing and retrieving of data. How does your data architecture meet the operational needs of AI systems and the strategic goals of the educational institution?

In addition to managing these key components, consider these practices:

- Develop and enforce data governance frameworks that align with your organisation's mission and goals.
- Establish teams or committees responsible for various aspects of data governance.
- Adapt organisational policies as AI technology evolves.
- Stay on top of legal requirements, like FERPA in the United States and GDPR in Europe.

Maintaining effective data governance in AI requires a blend of technical expertise, foresight and leadership to ensure your organisation remains compliant and leverages its data.

Copilot prompt

Examine the overlooked elements in implementing data governance in public educational institutions. Address the roles of stakeholders like IT leaders, administrators and educators and the challenges with types of data such as student performance and privacy. Provide actionable insights for improved governance.

Data privacy considerations in AI-driven education

Safeguarding students' personal data is paramount, especially as many AI applications – particularly those that are data-driven or use machine learning – require access to large datasets to learn, adapt and provide personalised experiences. The challenge is to balance the functionality and benefits of these AI applications with the need to protect sensitive data. Data privacy helps ensure individuals have control over their personal data's collection, usage and sharing. Review these key considerations.

Student, educator and faculty data privacy

Integrating AI in education requires careful management of both student and faculty data, including academic performance, learning behaviours and sensitive personal details. Safeguarding the privacy of this data is crucial to preventing breaches that could compromise the privacy of students and educators and harm your institution's reputation.

Compliance

Educational institutions are bound by laws and regulations like the Family Educational Rights and Privacy Act (FERPA) in the U.S., the General Data Protection Regulation (GDPR) in the EU and other regional data protection laws. Adhering to these regulations while implementing AI systems is a complex, but necessary task.

Practical tips

- Collect and use only the minimum data needed for the task. Limiting data collection reduces the risk of harmful data breaches.
- Where possible, anonymise student data to protect student identities by removing personally identifiable information (PII) or replacing it with pseudonyms.
- Have a plan in place for responding to data breaches, including steps for identifying and containing the breach.

Our commitment to collaboration

Microsoft is committed to fostering innovation in education through collaboration with leading edtech partners. This collaboration empowers organisations with advanced AI-driven tools to enhance teaching, learning and administrative efficiency. By integrating partner solutions with Microsoft's AI technologies, such as Microsoft Copilot, Azure AI Foundry and Microsoft 365 Copilot, these partnerships are designed to deliver tailored, scalable solutions for faculty, staff and administrators addressing the unique needs of their organisations. Together, Microsoft and its partners strive to create inclusive, accessible learning environments that prepare students for the future.

These collaborations go beyond technology integration, adopting a holistic approach that includes training resources, workshops, ongoing technical support and sharing best practices for trustworthy AI use in education. By working closely with edtech partners, Microsoft ensures that education leaders have access to a robust ecosystem of tools and expertise to successfully implement AI in their strategic planning and day-to-day operations. This collaborative approach accelerates AI adoption in education while reinforcing Microsoft's mission to empower every student and educator to achieve more.

Benefits for educational institutions

Partnering with edtech providers offers K-20 education institutions valuable benefits, enhancing their ability to leverage cutting-edge technologies and innovative solutions tailored to their specific needs. Through these collaborations, schools, colleges and universities gain access to specialised expertise and resources that streamline the integration of AI tools into their educational frameworks. Partners bring valuable insights and best practices from diverse implementations, enabling institutions to adopt AI more effectively and efficiently.

Partnerships enhancing AI integration in education

To support education leaders in harnessing AI's potential, Microsoft collaborates with leading edtech partners to deliver customised solutions addressing key challenges in education. The table highlights some of these partnerships, illustrating how each partner's offerings benefit educational institutions and align with Microsoft's AI solutions to foster innovative learning environments.

Through these collaborations, education institutions gain not only the tools and training needed to integrate AI, but also the confidence to lead the way. By empowering educators and administrators to maximise AI's potential, Microsoft and its partners are helping educational institutions be more innovative, inclusive and prepared to meet the needs of their students and institutions.

Partner	Benefits to educational institutions	Microsoft AI solution
Khanmigo for Teachers Khan Academy's AI-powered tutor for teachers	• Personalises learning support • Offers instant feedback for students • Reduces grading workload	Microsoft 365 Copilot
Kahoot! Learning product suite	• Saves time for educators • Improves search and brainstorming functionality • Creates quizzes and presentations on any topic	Azure OpenAI Service
Quizlet AI-powered study tools and flashcards	• Improves student retention • Offers personalised study paths • Engages students with interactive content	Microsoft 365 Copilot
DreamBox Learning Adaptive maths and reading programs	• Tailors learning to individual student needs • Provides real-time progress tracking • Supports differentiated instruction	Azure AI Studio
Canvas by Instructure Learning management system integration	• Fully immersive Teams meetings through LTI • OneDrive LTI support • Course roster sync in Teams through Class Teams LTI	Learning Tools Interoperability (LTI) using OneDrive LTI, Teams Meetings LTI and Teams Classes LTI
Schoology by PowerSchool Learning management system integration	• Fully immersive Teams meetings through LTI • OneDrive LTI support	Learning Tools Interoperability (LTI) using OneDrive LTI, Teams Meetings LTI and Teams Classes LTI
Blackboard by Anthology Learning management system integration	• OneDrive LTI support	Learning Tools Interoperability (LTI) using OneDrive LTI